

Class Code

May 26, 2026 5:03 PM

Starting slate:

```
MainActivity.kt
package com.example.week3demoLab

import ...

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) { //onCreate is mandatory lifecycle method
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView()
    }
}
```

**IMPORTANT* you need this life cycle state*

**important* anything you want in the UI needs to be referenced here.*

Added the following composable function:

```
@Composable
fun MyApp() {
    var count by remember { mutableStateOf(0) }
    Column(modifier = Modifier.padding(16.dp)) {
        Text(text = "Counter: $count", style = MaterialTheme.typography.bodyLarge)
        Spacer(modifier = Modifier.height(16.dp))
        Button(onClick = { count++ }) {
            Text("Increment")
        }
    }
}
```

denotes a composable function

variable, can be defined later

need to use modifier when doing padding, margins, etc.

□ Explanation of state management and composables.

- had to import everything manually
- for button: gave 3 options - always choose the composable one

```
// composable functions are outside main activity always
@Composable
fun MyApp() {
    var count by remember { mutableStateOf( value = 0) }
    Column(modifier = Modifier.padding( all = 16.dp)) {
        Text(text = "Counter: $count", style = MaterialTheme.typography.bodyLarge)
        Spacer(modifier = Modifier.height( height = 16.dp))
        Button(onClick = { count++ }) {
            Text( text = "Increment")
        }
    }
}
```

5:11
Counter: 20

Increment

counter increases every time button is clicked

```

74 // adds stylized text:
75 @Composable
76 fun StyledText() {
77     Text(
78         text = "Hello, Compose!",
79         //adds padding around the text
80         modifier = Modifier.padding( all = 24.dp),
81         style = TextStyle(
82             color = Color.Blue,
83             fontSize = 18.sp,
84             fontWeight = FontWeight.Bold
85         )
86     )
87 }

```

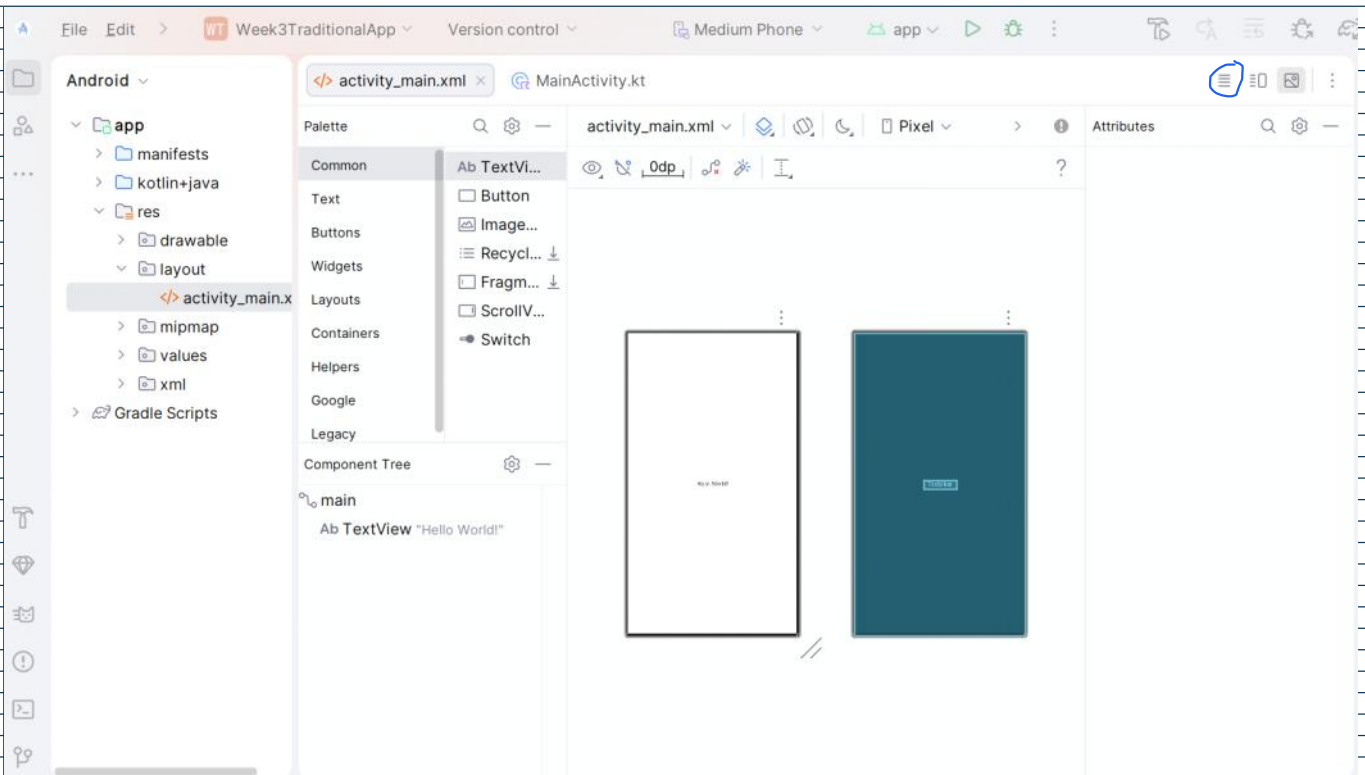
```

89 // COMPOSABLE LAYOUTS
90
91 // column layout:
92 @Composable
93 fun GreetingColumn() {
94     // stack Text elements vertically
95     Column {
96         Text(text = "Hello, Jetpack Compose!")
97         Text(text = "This is the second line.")
98         Text(text = "And here's the third.")
99     }
100 }
101
102 // row layout - 2 buttons
103 @Composable
104 fun ButtonRow() {
105     Row { // arrange the buttons horizontally
106         Button(onClick = { /* Handle click */ }) {
107             Text( text = "Button 1") // define button text here
108         }
109         Button(onClick = { /* Handle click */ }) {
110             Text( text = "Button 2")
111         }
112     }
113 }
114
115 // box layout
116 @Composable
117 fun IconTextOverlay() {
118     // create a container with a light gray background
119     Box(
120         modifier = Modifier.size( size = 150.dp).background(color = Color.LightGray).padding( all = 10.dp),
121         contentAlignment = Alignment.Center
122     ) {
123         Text( text = "Look here!", Modifier.align(Alignment.Center))
124         // Icon component aligned to the top end of the Box, overlapping the text
125         Icon(
126             imageVector = Icons.Filled.Favorite,
127             contentDescription = "Favorite",
128             modifier = Modifier.align(Alignment.TopEnd).size( size = 40.dp)
129         )
130     }
131 }
132
133

```

Made new project w/ Empty Views Activity to see TRADITIONAL way of building Uis

- Layout is constraint layout
 - o have to add A LOT of properties
 - o supports drag & drop



```
activity_main.xml MainActivity.kt
1 package com.example.week3traditionalapp
2
3 > import ...
4 // language is still Kotlin
5 2 Usages
6
7 class MainActivity : AppCompatActivity() {
8     override fun onCreate(savedInstanceState: Bundle?) {
9         super.onCreate(savedInstanceState)
10        enableEdgeToEdge()
11        setContentView(R.layout.activity_main) // add an xml view FILE (in resources > layout)
12        ViewCompat.setOnApplyWindowInsetsListener(view = findViewById(id = R.id.main)) { v, insets -
13            val systemBars = insets.getInsets(typeMask = WindowInsetsCompat.Type.systemBars())
14            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
15            insets
16        }
17    }
18 }
19
20 }
```

Traditional MainActivity

```
src > main > java > com > example > week3traditionalapp > MainActivity
DemoLab Version control Medium Phone app
MainActivity.kt
7 import androidx.compose.ui.unit.dp
8 import androidx.compose.ui.unit.sp
9 import com.example.week3demolab.ui.theme.Week3DemoLabTheme
10
11 1 Usage
12
13 class MainActivity : ComponentActivity() {
14     override fun onCreate(savedInstanceState: Bundle?) { //onCreate is mandatory lifecycle method
15         super.onCreate(savedInstanceState)
16         enableEdgeToEdge()
17         setContent {
18             // call composable functions here
19             // MyApp() // < that's it!
20             // StyledText()
21             // GreetingColumn()
22             // ButtonRow()
23             IconTextOverlay()
24         }
25     }
26 }
```

Jetpack Compose Main Activity